EXP NO: 4 EXPERIMENT TO UNDERSTAND THE DATA PREPROCESSING IN DATA SCIENCE

AIM:

Understand the importance of Data preprocessing in data science

DATASET:

Country	Age	Salary	Purchased
France	44	72000	No
Spain	27	48000	Yes
Germany	30	54000	No
Spain	38	61000	No
Germany	40		Yes
France	35	58000	Yes
Spain		52000	No
France	48	79000	Yes
Germany	50	83000	No
France	37	67000	Yes

CODE:

```
import numpy as np
import pandas as pd

# Create DataFrame directly
data = {
    "Country": [
        "France",
        "Spain",
        "Germany",
        "Spain",
        "Germany",
        "France",
        "Spain",
        "France",
        "Germany",
        "France",
    ],
    "Age": [44, 27, 30, 38, 40, 35, np.nan, 48, 50, 37],
    "Salary": [
```

```
        72000,
        48000,
        54000,
        61000,
        np.nan,
        58000,
        52000,
        79000,
        83000,
        67000,
    ],
    "Purchased": [
        "No",
        "Yes",
        "No",
        "No",
        "Yes",
        "Yes",
        "No",
        "Yes",
        "No",
        "Yes",
    ],
}
df = pd.DataFrame(data)
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
df = pd.get_dummies(df, columns=["Country"], dtype=int)
df["Purchased"] = np.where(df["Purchased"] == "Yes", 1, 0)
df["Age"] = (df["Age"] - np.mean(df["Age"])) / np.std(df["Age"])
df["Salary"] = (df["Salary"] - np.mean(df["Salary"])) / np.std(df["Salary"])

print(df)
```

OUTPUT:

```
        Age        Salary  Purchased  Country_France  Country_Germany  \
0  0.758874  7.494733e-01          0               1                0
1 -1.711504 -1.438178e+00          1               0                0
2 -1.275555 -8.912655e-01          0               0                1
3 -0.113024 -2.532004e-01          0               0                0
4  0.177609  6.632192e-16          1               0                1
5 -0.548973 -5.266569e-01          1               1                0
6  0.000000 -1.073570e+00          0               0                0
7  1.340140  1.387538e+00          1               1                0
8  1.630773  1.752147e+00          0               0                1
9 -0.258340  2.937125e-01          1               1                0

   Country_Spain
0              0
1              1
2              0
3              1
4              0
5              0
6              1
7              0
8              0
9              0
```

RESULT:

Thus the code is executed successfully.